NMC 2013 A-LEVEL PAPER 2 QUESTIONS

SECTION A : 5 marks each

1. The average of 16 **different** positive integers is 16. What is the second greatest possible value that any of these integers could have?

2. A security guard works for four days consecutivaly, then has the next day off, works four more days, has a day off , and so on. Today is a sunday and a day off. On how many days does the guard work before next having a day off on a Sunday?

3. If a is non zero and the three roots of $ax^3 + bx^2 + (a-4)x + 2 = 0$ are equal integers, with r the common value of these integers. Determine the value of r.

4. One Half the books on a teacher's bookshelf are mathematics books, a third of them are physics books, and $\frac{1}{15}$th of them are history books. The remainder of the books are romance novels. If 2 of the mathematics books, and 4 of the physics books are replaced by romance novels, then romance novels will comprise 15% of the books on the bookshelf. How many books, total, are there on the bookshelf?

5. If the equation $x^4 - ax^3 + 2x^2 - bx + 1 = 0$ has a real root. Show that $a^2 + b^2 \geq 8$.

6. Let x and y be real numbers. Find the maximum value of $x - y$ if $2(x^2 + y^2) = x + y$.

7. The real numbers x and y satisfy the equations $xy = sin(2x)$ and $\frac{x}{y} = tan(t)$ where $0 < t < \frac{\pi}{2}$.

   What is the value of $x^2 + y^2$?

8. Show that if $|x| < 1$, $|y| < 1$, then $|x - y| < 1 - xy$.

9. The five-digit number $a679b$ where a and b are digits, is divisible by 36. Find all possible such five-digit numbers.

10. On Monday, Ruth tells the truth for the whole day, otherwise he lies for the whole day. Today is Monday and Ruth has made exactly FOUR of the following statements. Which statement could he have not made today?
    A. I have a prime number of friends. B. I have as many male friends as female friends
    C. My name is Charles. D. I always speak the truth. E. Three of my friends are older than me. SA PAGE 50

SECTION B : 10 marks each

11. Immaculate is fascinated by triangle numbers $(1, 3, 6, 10, 15, 21,$ etc) and recently coming across a clock, he found that he could rearrange the twelve numbers $1, 2, 3, \cdots, 12$ around the face so that each digit pair added up to a triangle number. He left 12 in its usual place. What number did he put where the 6 would usually be?

12. An "unfortunate" number is a positive integer which is equal to 13 times the sum of its digits. Find all "unfortunate" numbers less than 1000?

13. The letters S, M and C represent whole numbers. If $S \times M \times C = 240$, $S \times C + M = 46$, $S + M \times C = 64$. What is the value of $S + M + C$?

14. If the square is completed with the letters A, B, C, D and E so that no row, column or either of the two main diagonal lines contains the same letter more than once,which letter should replace the asterisk?

	*			A
		B		
D		C		
			E	

15. In the trapezium shown(not to scale), XY is parallel to two sides and passes through the point of intersection of the diagonals. What is the length XY?

NMC 2013 A-LEVEL PAPER 2 SOLUTIONS

SECTION A

1. $xy = \sin(2x)$ and $\frac{x}{y} = \tan t \Longrightarrow x^2 + y^2 = y^2\tan^2 t + y^2 = y^2(\tan^2 t + 1) = y^2\sec^2 t$

   $= xy\frac{y}{x}\sec^2 t = \frac{\sin(2t)\cos t}{\tan t} = \frac{2\sin t\cos t}{\sin t\cos t} = 2.$

2. In order to make one number the largest possible, the other fifteen numbers should be the smallest possible numbers.
   Let y be the largest number possible, then

$$\frac{1+2+3+4+5+6+7+8+9+10+11+12+13+14+15}{16} = 16$$

   $120 + y = 256 \Longrightarrow y = 256 - 120 = 136.$

3. The number of work days until the guard's new Sunday off is a multiple of 7 and 5. The smallest multiple of 7 and 5 is 35, hence the number of work days before the guard's new Sunday off is 35 days of which there are 28 work days.

4. Denote Florence's age by ab. Suppose that $b \geq 6$, then $ab = a(b-6)$ which equation yields $a = 0$ contradicting the assumption that a is non zero. This implies that $b \leq 5$ and $ab = (a-1)(b+4)$ or $4a - b = 4$. The only non zero solution is $a = 2$ and $b = 4$. We therefore have that Florence's age is currently 24 and the product of the digits of the current age is 18.

5. Given the equation $x^4 - ax^3 + 2x^2 - bx + 1 = 0$. Let y be the root of the equation. Then

$$y^4 - ay^3 + 2y^2 - by + 1 = 0 \text{ and so } y^2 - ay + 2 - \frac{b}{y} + \frac{1}{y^2} = 0$$

$$\text{Completing the squares gives } (y - \frac{a}{2})^2 + (\frac{1}{y} - \frac{b}{2})^2 - \frac{a^2}{4} - \frac{b^2}{4} + 2 = 0$$

$$(y - \frac{a}{2})^2 + (\frac{1}{y} - \frac{b}{2})^2 = \frac{a^2}{4} - \frac{b^2}{4} + 2 \geq 0$$

$$\frac{a^2 + b^2 - 8}{4} \geq 0 \Longrightarrow a^2 + b^2 \geq 8.$$

6. Let x denote the girl-boy sets, then there are $(x + 4)$ girl-girl sets. Therefore there are $35 - (x + x + 4) = 31 - 2x$ boy-boy sets, and so
   $x + 2(31 - 2x) = 38 \Longrightarrow x + 62 - 4x = 38 \Longrightarrow 3x = 24$ so that $x = 8$. Thus, there are 8 girl-boy sets and the boy-boy sets of twins are therefore $31 - 16 = 15$.

7. The positive odd integers are grouped into sets of 3 as follows: Set 1 : $\{1, 3, 5\}$; Set 2 : $\{7, 9, 11\}$; Set 3 : $\{13, 15, 17\}$.
   Recall that odd numbers are always denoted by $2n - 1$ where $n \in \mathbb{N}$.
   By using positions to locate required values, Set 1 ends with the third odd number 5, Set 2 ends with the sixth odd number 11 and Set 3 ends with the ninth odd number 17. Therefore, the position of the last odd numbers is given by $\{3^{rd}, \{6^{th}, \{9^{th}, \cdots\}$ in the $\{1^{st}, \{2^{nd}, \{3^{rd}, \cdots\}$ positions respectively. Therefore the last odd number in the 2013th set is the $3(2013)^{th}$ odd number or 6039th odd number. Now since $2 \times 6039 - 1 = 12077$. The set 2013 required is $\{12073, 12075, 12077\}$.

8. Since $a679b$ is divisible by 36, it is also divisible by 4 and 9. If a number is divisible by 4, then the number formed by the last 2 digits must be a multiple of 4. Hence 4 divides $9b$ and so $b = 2$ or $b = 6$.
   Similarly, if a number is divisible by 9, the sum of the digits is a multiple of 9 and hence 9 divides $a + 6 + 7 + 9 + b = a + b + 22$.
   Now, when $b = 2$, it implies $a = 3$ because 9 divides $a + 24$ and when $b = 6$, it implies $a = 6$ because 9 divides $a + 28$. Therefore the possible numbers are 36792 and 86796.

9. For $n > 5$, we have $1 < \frac{n+3}{n-1} < 2$ and for $n < -3$, we have $0 < \frac{n+3}{n-1} < 1$ and so we need to consider nine integer values of n from -3 to $+5$ only.

   We find that $\frac{n+3}{n-1}$ is an integer except for $n = -2, n = 1$ and $n = 4$ giving six acceptable values of n.
   Alternatively,
   We can write $\frac{n+3}{n-1}$ as $\frac{n+3}{n-1} = \frac{n-1+4}{n-1} = 1 + \frac{4}{n-1}$. The number on the far left is an integer if and

   only if $\frac{4}{n-1}$ is an integer. Since the divisors of 4 are $1, 2$ and 4, the only possibilities are $n - 1 = \pm 1, \pm 2, \pm 3$ so that $n = 0, 2, -1, 3, -3, 5$.

10. Statement C is true and Statement D is false. There are two possibilities. Either Ruth is telling the truth today and does not does not make Statement D or she is lying and does not make Statement C. Suppose A, B and D are true, from statement E, Ruth has atleast 3 friends and from A, this number of friends is prime and from B the number of her friends is even. The only even prime number is 2 so these statements cannot all be true and hence Ruth cannot be telling the truth today. However, it is possible that all the 3 statements are false so Ruth is lying today and could not have made statement C.

SECTION B

11. Multiply $S + M \times C = 64$ by S to get $S^2 + S \times M \times C = 64S \Longrightarrow S^2 + 240 = 64S$ and solving this equation gives $S = 60$ or $S = 4$.

    Multiply $S \times C + M = 46$ by M to get $M^2 + S \times M \times C = 46S \Longrightarrow M^2 + 240 = 64M$ and solving this equation gives $M = 40$ or $M = 6$.

(i) If $M = 40$, this implies $S \times C \times 40 = 240 \Longrightarrow S \times C = 6$. Either $C = \frac{6}{40}$ or $C = \frac{6}{4}$ because the value of S is either 60 or 4.

(ii) If $M = 6$, this implies $S \times C \times 60 = 240 \Longrightarrow S \times C = 40$. Either $C = \frac{40}{60}$ or $C = \frac{40}{10} = 4$. Hence the only whole number solutions of the equations are $S = 4, M = 6$ and $C = 10$.

12. Let the vertices of the trapezium be A, B, C, D and $\overline{AC}$ meet $\overline{BD}$ at E as shown below. Triangle

ABE and CDE are similar since $< ABE =< CDE$ (alternate angles) and $< AEB =< CED$(vertically opposite angles). This implies $\frac{CE}{AE} = \frac{CD}{AB} = \frac{18}{3} = 3$.
Hence $AC = 4AE$ and by a similar argument $BD = 4BE$.
Triangles AXE and ADC are similar since $< XAE =< DAC$ and $< AXE =< ADC$ (corresponding angles), so that $\frac{XE}{DC} = \frac{AE}{DC} = \frac{1}{4}$. Hence $XE = 4.5cm$.
Applying similar argument to triangles BYE and BCD, we find that $YE = 4.5cm$ also. Hence the length $XY = 4.5 + 4.5 = 9cm$.

13. (i) Clearly, there is no single digit magic number.

(ii) Let ab be a two digit magic number, Then $10a + b = 13(a + b) \Longrightarrow 3a + 12b = 0$. This is an impossibility since both a and b are greater than zero. Hence there are no two digit magic numbers.

(iii) Let abc be a three digit magic number, Then $100a+10b+c = 13(a+b+c) \Longrightarrow 29a = b+4c$. The only possible value of a is 1 since otherwise $29a \geq 58$ and yet the maximum value of $b + 4c$ can only be 45. The equation $29a = b + 4c$ has exactly non negative integers $(b, c) = (1, 7), (5, 6)$ and $(9, 5)$.
Hence $117, 156, 195$ are the only magic numbers less than 1000.

14. The equation $ax^3 + bx^2 + (a - 4)x + 2 = 0 \Longrightarrow x^3 + \frac{b}{a}x^2 + \frac{a-4}{a}x + \frac{2}{a} = 0$.
Sum of roots, $3r = \frac{b}{a}$ and product of roots $r^3 = \frac{-2}{a} \cdots\cdots\cdots (i)$.
Since the equation has repeated roots, $x = r$ is also a solution to $f'(x) = 0$. Therefore
$3ax^2 + 2bx + (a - 4) = 0$ so that the product of roots $r^2 = \frac{a-4}{3a} \cdots\cdots\cdots (ii)$. It follows from (i) and (ii) that $3r^2 = \frac{a-4}{a}$ and $-r = \frac{2}{a}$. Thus
$r(\frac{a-4}{3a}) = r^3 = -\frac{2}{a} \Longrightarrow r = \frac{6}{4-a}$, so that $3(\frac{6}{4-a})^2 = \frac{a-4}{a}$ which simplifies to the cubic equation $(a - 4)^3 - 108a = 0$ with roots $a = 2$ and $a = 16$.
This gives $r = 1$ or $r = -\frac{1}{2}$.

15.

	$*$			A
		B		
D	E_2	C	A_1	B_3
			E	

The square labelled A_1 is in the same row as C and D, the same column as E and the same diagonal as B, so it must be A. The square labelled E_2 is the next to be filled in as is in the same row as A, C and D and in the same diagonal as B. The square labelled B_3 now completes the fourth row. The bottom left-hand corner must now be A.This is because one of the three remaining squares in the diagonal which runs from bottom left to top right must be A, but A at the end of row 2 means that A cannot be in either of the last 2 squares of this diagonal. The square marked $*$ is in the same diagonal as A, B and C and in the same column as E so it must be D.